

Project Research Documentation

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1. Project Title

Email Writer Assistant - Multimodal RAG Based Real-World Email Assistant

2. Problem Statement

Many users face difficulties while writing professional emails due to:

- Lack of knowledge about professional email formats.
 - Difficulty choosing the appropriate tone (Formal, Business, Friendly, Apology, etc.).
 - Time-consuming process of reading documents and manually writing emails.
 - Inability of existing tools to effectively utilize information from PDFs, images, audio, and videos.
 - Need to summarize multiple documents before drafting an email.
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3. Proposed Solution

Develop a Multimodal RAG-Based AI Email Writer Assistant that:

- Accepts **Text, PDF, Image, Audio, and Video** as input.
 - Extracts and understands relevant information from uploaded content.
 - Uses a Retrieval-Augmented Generation (RAG) pipeline to retrieve context.
 - Generates professional, personalized emails with appropriate subject lines and tone.
 - Saves users' time while improving communication quality and productivity.
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4. Domain

This project belongs to the following technology domains:

Domain	Contribution to the Project
Generative AI	Generates professional and personalized email content based on user input.
Natural Language Processing (NLP)	Understands user prompts, extracts information, and generates human-like email text.
Retrieval-Augmented Generation (RAG)	Retrieves relevant information from uploaded documents before generating the email, improving accuracy and personalization.
Multimodal AI	Processes multiple input types such as Text, PDF, Images, Audio, and Video.
Productivity Tools	Helps users save time, improve communication, and automate professional email writing.

5. Project Features

Core Features:

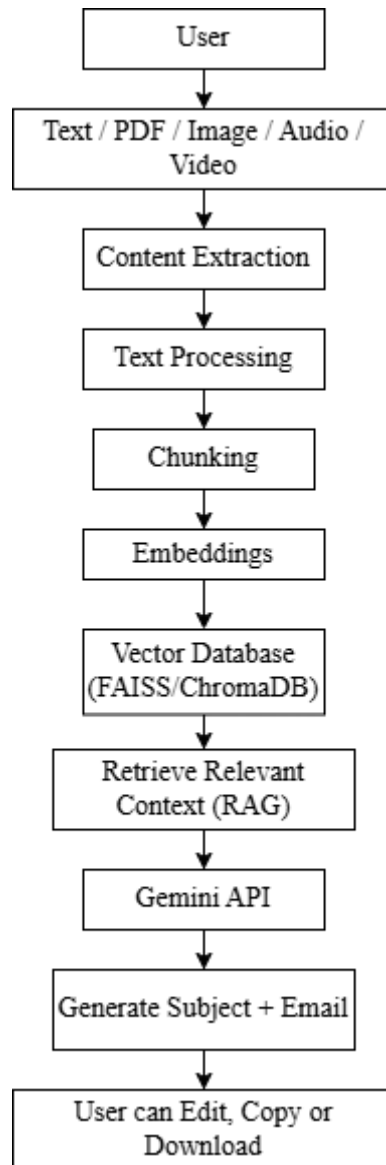
Feature	Description
AI Email Generation	Generates professional and personalized emails based on user input.
Multimodal Input Support	Accepts Text, PDF, Images, Audio, and Video as input.
RAG-Based Context Retrieval	Retrieves relevant information from uploaded documents before generating emails.
Automatic Subject Line Generation	Creates an appropriate subject line based on the email content.
Multiple Email Tones	Supports Formal, Business, Friendly, Apology, Request, and Follow-up email styles.
Grammar & Writing Improvement	Enhances grammar, clarity, and professionalism of generated emails.
Email Editing	Allows users to edit the generated email before copying or downloading.

Feature	Description
Copy & Download	Enables users to copy or download the generated email easily.

Advanced Features:

Feature	Description
Resume-Based Job Application Emails	Generates personalized job application emails using uploaded resumes.
PDF Understanding	Extracts and understands information from PDF documents.
Image Understanding	Reads information from certificates, invitations, screenshots, and other images.
Audio-to-Email	Converts voice recordings into professional emails.
Video Summarization	Extracts important information from videos to generate emails.
Smart Tone Recommendation	Suggests the most suitable email tone based on the user's purpose.
Multilingual Email Generation	Generates emails in different languages.
Email Length Control	Allows users to generate Short, Medium, or Detailed emails.
AI Follow-up Suggestions	Suggests follow-up emails after the original email.
History Management	Saves previously generated emails for future reference.
Dark Mode & Responsive UI	Provides a modern interface that works across different devices.

6. System Workflow / Project Architecture



7. Technology Stack

1. Frontend

- React.js

2. Backend

- Python
- FastAPI

3. AI & RAG

- Gemini API
- LangChain
- Embedding Model
- FAISS / ChromaDB

4. Document & Media Processing

- PyPDF
- OpenCV
- OCR
- Whisper
- MoviePy / FFmpeg

5. Development & Version Control

- Git
- GitHub

8. Data Collection Strategy

The Email Writer Assistant requires various types of data to generate accurate, professional, and context-aware emails. The following data will be collected and processed:

Data Type	Purpose
Professional Email Samples	Learn professional email structure and writing style.
Job Application Emails	Generate personalized job application emails.
Leave Request Emails	Create formal leave request emails.
Business Emails	Support official and workplace communication.
Complaint & Apology Emails	Generate polite complaint and apology emails.
Follow-up Emails	Assist users in writing professional follow-up messages.
Resumes	Extract candidate information for job-related emails.
College & Academic Documents	Generate emails related to admissions, internships, and academics.
PDF Documents	Retrieve relevant information through the RAG pipeline.
Images	Extract text and information using OCR.
Audio Files	Convert speech into text for email generation.

Data Type	Purpose
Video Files	Extract audio and summarize relevant information for email generation.

Data Sources

- Public email datasets
- Kaggle datasets
- Official sample letters and email formats
- Publicly available PDF documents
- Sample resumes and academic documents
- User-uploaded files (processed only during email generation)

9. Project Interface (UI)

The Email Writer Assistant is designed with a clean, user-friendly, and intuitive interface that allows users to generate professional emails with minimal effort.

Main Interface Components

1. Navigation Sidebar

- Home
- New Email
- Email History
- Settings
- Help

2. Chat Workspace

- AI conversation area
- User prompt input box
- AI response display

3. File Upload Panel

Users can upload:

- PDF Documents
- Images
- Audio Files
- Video Files

4. Email Configuration

- Email Tone Selection
- Email Length Selection
- Language Selection

5. Generated Email Preview

Displays:

- Subject Line
- Email Body

- Attachment Summary (if applicable)

6. Action Buttons

- Send
- Regenerate
- Copy
- Download

User Flow

1. Enter a prompt or upload files.
2. Choose the email tone and preferences.
3. Click Generate Email.
4. Review the generated subject and email body.
5. Edit if required.
6. Copy or download the final email.

Interface Design

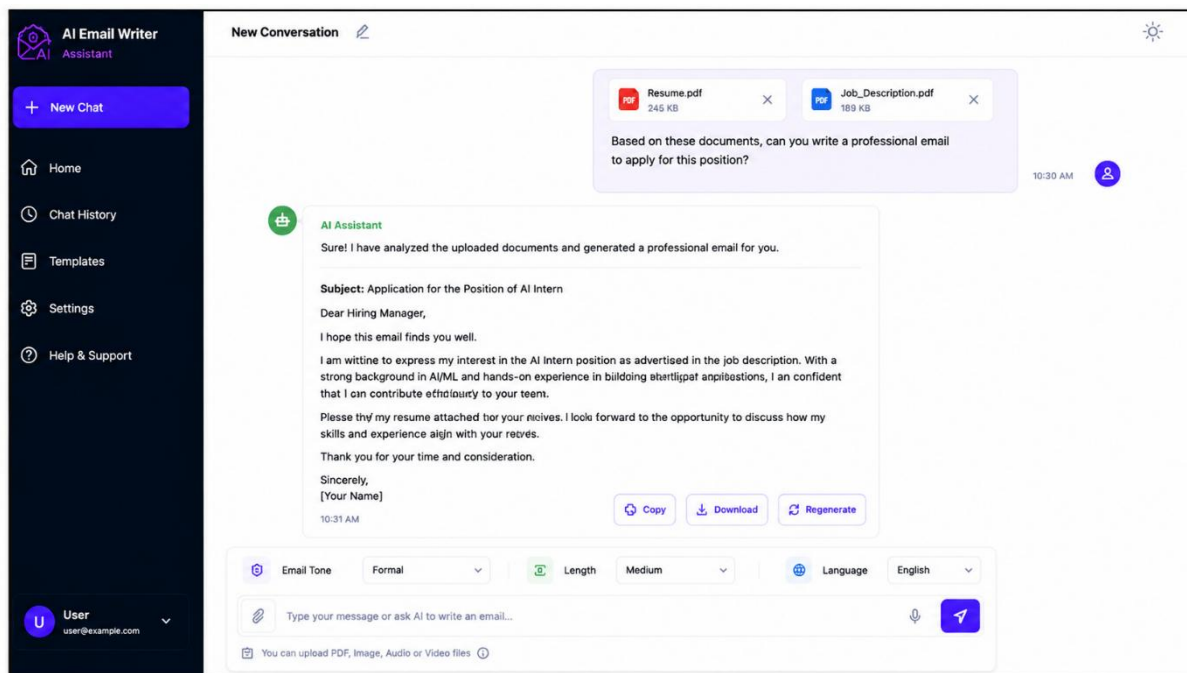


Fig. 1: Proposed User Interface of the Email Writer Assistant

10. Expected Output

The Email Writer Assistant will generate a professional and context-aware email based on the user's prompt and uploaded files. The final output will include:

- A suitable email subject line.
- A well-structured professional email body.
- Support for different writing tones (Formal, Business, Friendly, Request, Apology, Follow-up, etc.).

- An editable email for further customization.
 - Options to Copy, Download, or Regenerate the generated email.
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11. References

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3. Google Search – Used to understand AI concepts, RAG, Multimodal AI, and related technologies.
4. Mailmodo – AI Email Writer (Reference for AI email writing features and user interface). [Mailmodo AI Email Writer](#)
5. Editpad – AI Email Writer (Reference for email generation workflow and interface ideas). [Editpad AI Email Writer](#)